Data Science Pipeline

Data science Week-3 DS_04

Name – Rushay Gopani And Ajay

About this document

This report documents three pieces of data science work carried out on NHS Talking Therapies (IAPT) data during Week 3. Each piece tackles a different question that NHS commissioners and planners regularly face:

- How many people are likely to be referred to talking therapies over the next few months?
- Are any NHS regions prescribing in ways that look unusual and might need a closer look?
- Which parts of England are most deprived but getting the least mental health support?

The report is written so that both clinical and non-technical readers can follow what was done, why certain choices were made, and what the results mean in practice. Code details are kept brief — the focus is on what the numbers tell us and where the analysis has limits.

All data used comes from publicly available NHS sources. No patient names or identifiable information was used at any point.

Part-1 – Forecasting Referral Demand

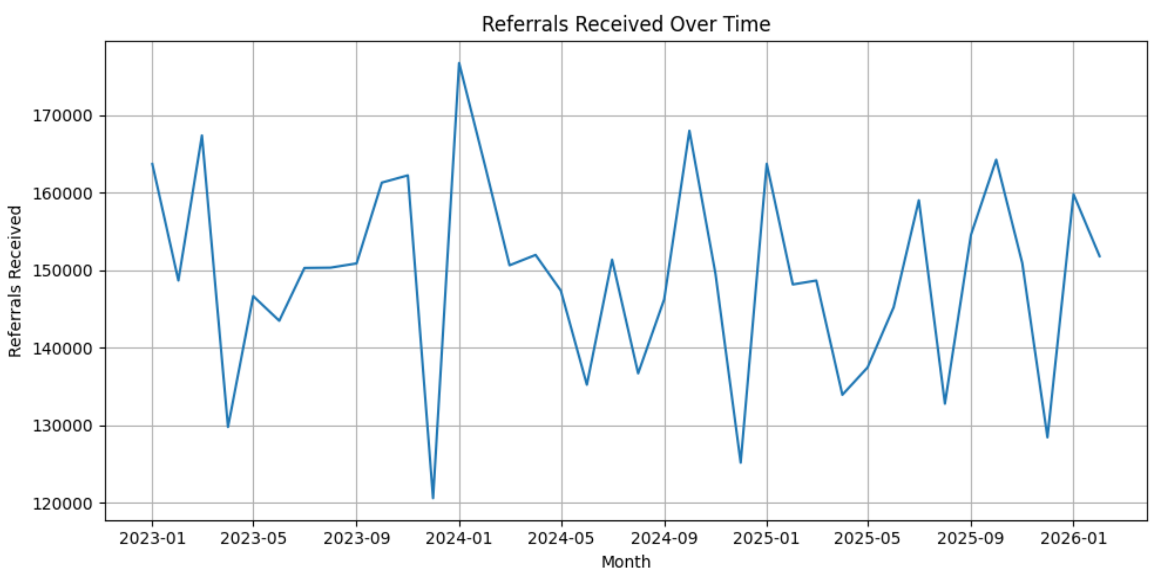
1) What problem does this solve?

NHS Talking Therapies services need to plan staff rotas, therapy slots, and waiting list management weeks or months in advance. Without a reasonable forecast of how many referrals are coming in, services either over-staff (wasting money) or under-staff (leaving patients waiting longer than they should).

This model uses past referral data to predict how many referrals are expected in the next three months, along with a confidence range so planners know how much uncertainty to build in.

2) Where did the data come from?

Monthly referral counts were pulled from the NHS Talking Therapies monthly statistics — specifically the Count_ReferralsReceived measure. The dataset covers January 2023 to February 2026, giving 38 monthly data points.



Looking at the chart, a few things stand out. Referrals dip sharply around November every year — likely reflecting fewer GP appointments over the holiday period — and spike in January. There is no obvious long-term upward or downward trend, which is useful: it means a forecasting model does not need to account for structural growth.

3) How was the model built?

The 38 months of data were split into two groups:

- Training data — the first 32 months (up to August 2025), used to teach the model the patterns in referrals
- Test data — the final 6 months (September 2025 to February 2026), used to check whether the predictions matched what actually happened

Two different forecasting methods were tried side by side so we could pick the better one.

Method 1 — Prophet

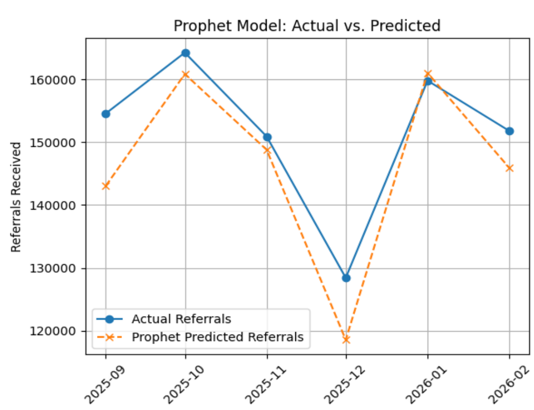
Prophet is an open-source forecasting tool originally developed by Meta. It is designed for the kind of messy, irregular time series that real-world data tends to look like. It automatically picks up on weekly, monthly, and yearly patterns without needing much manual configuration.

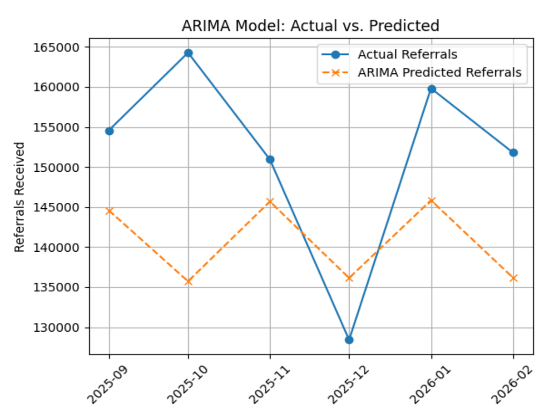
Method 2 — ARIMA

ARIMA (AutoRegressive Integrated Moving Average) is a classical statistical method that has been used in forecasting for decades. It works by looking at how past values relate to the current value, and how recent errors influence the next prediction. The version used here was ARIMA(2,1,2).

4) Which model performed better?

Measure	Prophet	ARIMA	Winner
MAE (lower is better)	5,679	13,518	**Prophet ✓**
RMSE (lower is better)	6,869	15,499	**Prophet ✓**



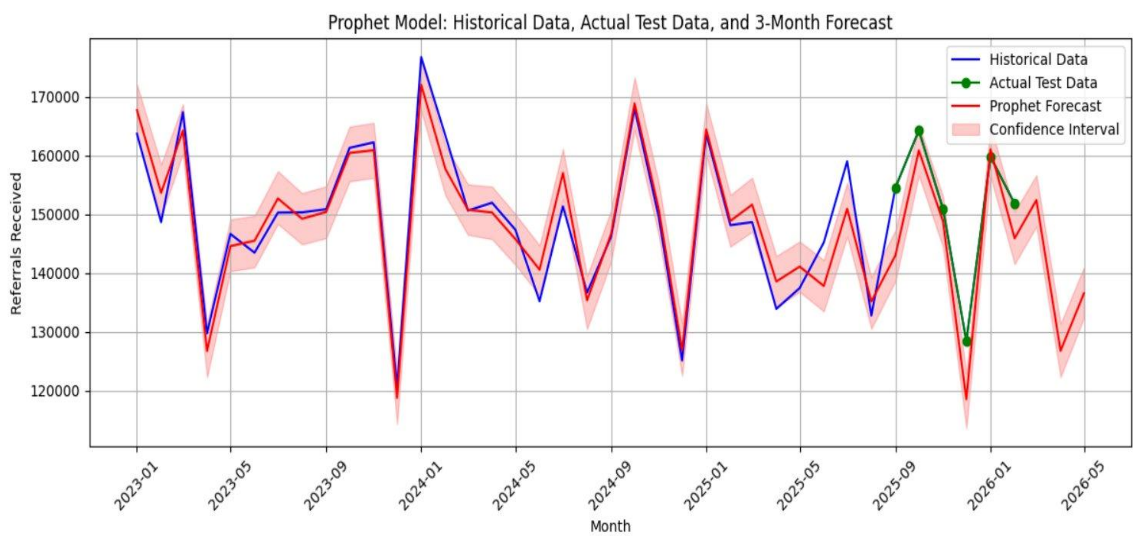


Prophet was the clear winner. Its average error (MAE) of around 5,679 referrals means it was off by about 3.5% on average — which is quite reasonable for a monthly health services forecast. ARIMA missed by more than double that, particularly struggling with the seasonal dips.

The gap is not surprising. ARIMA assumes the statistical relationship between months stays constant, which is a fairly rigid assumption. Prophet handles changing seasonal patterns more flexibly, which suits the ups and downs in IAPT referral volumes.

5) What does the forecast say?

Month	Predicted referrals	Low estimate	High estimate
March 2026	152,424	147,954	156,682
April 2026	126,767	122,320	131,351
May 2026	136,547	132,375	140,986



April is expected to see a noticeable dip to around 127,000 referrals before recovering in May. This follows the pattern seen in previous years and likely reflects fewer working days and different GP consultation patterns around Easter. Services should plan accordingly and not interpret this seasonal dip as a structural reduction in demand.

6) Assumptions and Dataset Issue

- The model assumes the patterns seen in 2023–2025 will broadly continue. A major policy change or a public health event would not be reflected.
- Only 38 months of data were available. More data would make the seasonal patterns easier to detect reliably.
- The model does not know about school holidays, bank holidays, or regional events. A version tuned with UK public holidays would likely reduce the error further.
- The confidence band widens as you go further out. The May figure is less certain than the March figure.
- This is an England-wide aggregate. Individual ICBs may see very different patterns.

Part 2 – Spotting Unusual Prescribing Patterns

1) What problem does this solve.

NHS prescribing data is huge — thousands of CCGs reporting every month across hundreds of drug types. Spotting which of those CCG-month combinations look genuinely unusual is nearly impossible to do by eye. Manual audits tend to focus on known problem areas rather than finding new ones.

This analysis uses a machine learning technique called Isolation Forest to automatically scan through prescribing records and flag the ones that look statistically unusual. The flagged records are not automatically "wrong" — they need clinical review — but they give auditors a focused shortlist rather than having to check everything.

2) Where did the data come from

Two datasets were downloaded from the OpenPrescribing API, which publishes NHS prescribing statistics for Clinical Commissioning Groups across England:

a. Spending by CCG — monthly totals of actual cost, number of prescription items, and total medicine quantity for each CCG (6,421 rows)
b. Organisation details — the registered patient population (list size) for each CCG each month (6,360 rows)

The two datasets were joined together on CCG code and date, giving a merged dataset of 6,360 CCG-month records.

One practical limitation: the OpenPrescribing API uses Cloudflare protection that blocks automated downloads. The data had to be downloaded manually via a web browser, which means this pipeline cannot yet be set up to refresh automatically each month.

3) Why not just look at total spend

A CCG with 500,000 registered patients will naturally spend far more on prescriptions than one with 100,000. Looking at raw totals would just flag the biggest CCGs every time, regardless of whether their prescribing behaviour is actually unusual.

To get around this, five population-adjusted metrics were calculated — all of which divide out the size effect:

Metric name	How it is calculated	Why it matters
Cost per patient	Total spend ÷ population	Removes the size effect — big CCGs naturally spend more
Items per patient	Prescriptions issued ÷ population	Shows how intensively a CCG prescribes per person
Quantity per patient	Total medicine volume ÷ population	Catches bulk prescribing patterns
Cost per item	Total spend ÷ prescriptions	Flags unusually expensive drugs
Quantity per item	Total volume ÷ prescriptions	Spots abnormally large pack sizes

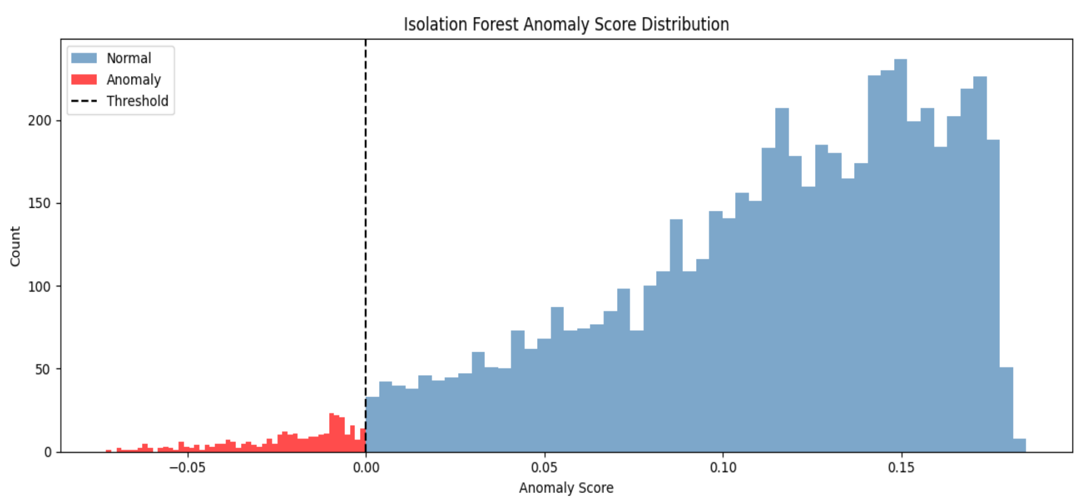
These five metrics were then standardised so they are all on the same scale before feeding them into the model. Without standardising, a metric measured in pounds would dominate one measured as a count of items.

4) How does Isolation Forest work

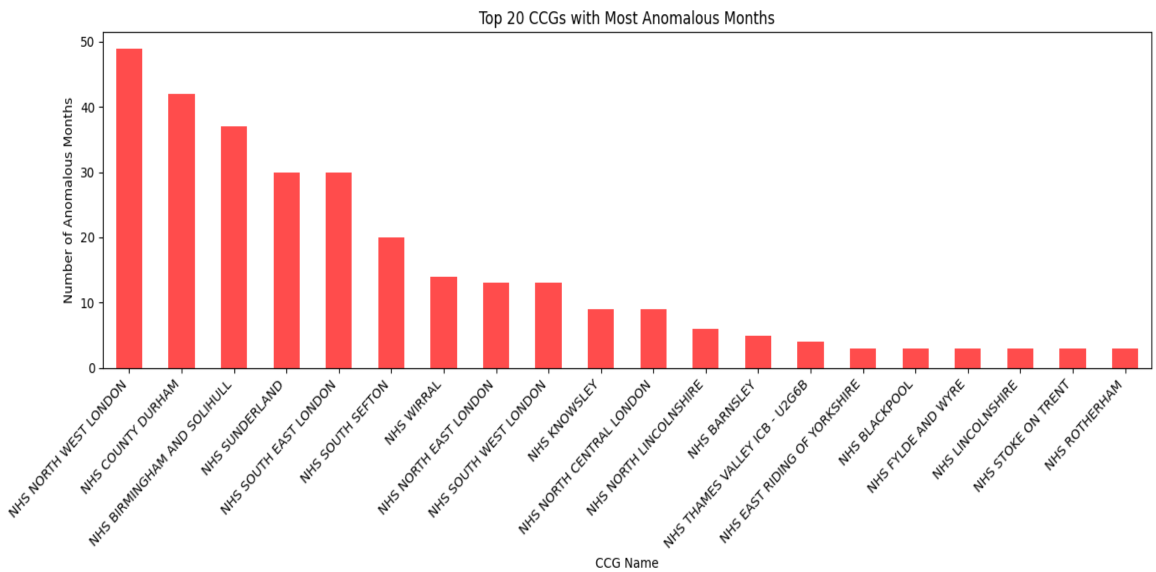
The Isolation Forest algorithm works on a simple but clever idea: if a data point is unusual, it is easy to isolate. The algorithm repeatedly slices the data with random cuts. Normal points, clustered in the middle of the distribution, need lots of cuts before they are isolated. Outliers, sitting far from the pack, get isolated quickly.

Each record gets an anomaly score. Scores below zero are flagged as anomalies. The model was set to expect around 5% of records to be anomalous (318 out of 6,360), which is a reasonable starting assumption for a prescribing audit context.

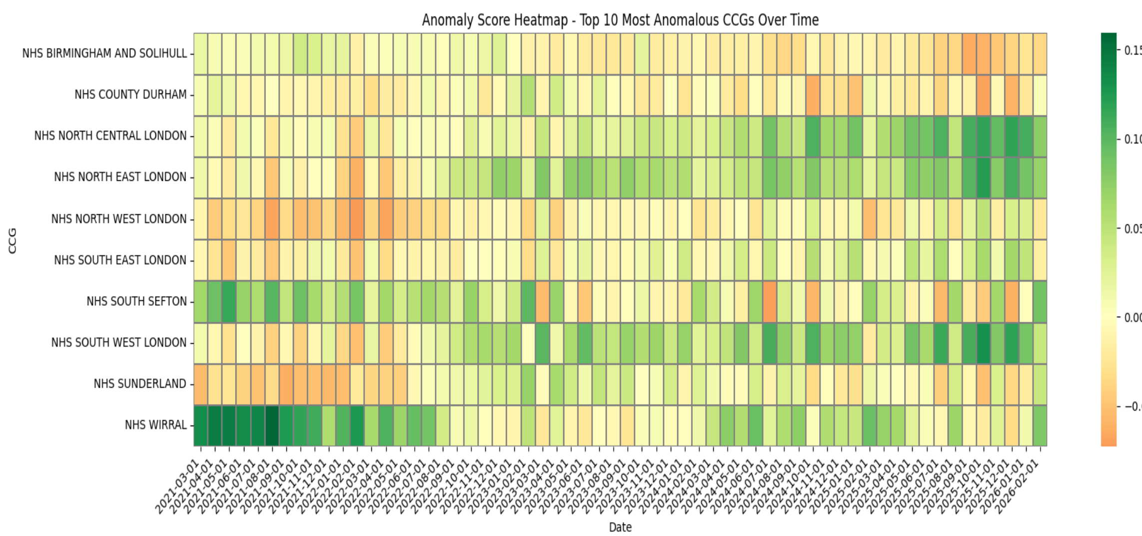
5) What did the model find



The separation between normal and anomalous records is fairly clean. Most flagged records have strongly negative scores, meaning the model is reasonably confident they are genuinely unusual rather than borderline cases.



Some CCGs show up as anomalous across many months in a row. That kind of persistent pattern is worth looking at more closely — it suggests the unusual behaviour is structural rather than a one-off data error.



The heatmap is particularly useful for a clinical reviewer because it shows timing. A CCG that was normal for two years and then suddenly became anomalous in mid-2023 is a very different situation from one that has been consistently flagged throughout.

Important to note: a flagged record does not mean something wrong happened. Anomalies can arise from legitimate reasons — a one-off bulk order, a boundary change merging two CCGs, or a genuine data entry issue. Each flagged record needs a human to look at it and make a judgement.

6) What gets handed over

The model produces a file called anomaly_flags.csv which contains every CCG-month record in the dataset, with two extra columns added: an anomaly score (continuous) and an anomaly label (Normal or Anomaly). This file is ready to be filtered and reviewed in Excel or any reporting tool.

7) Assumptions and honest review

- The 5% contamination setting is a starting assumption, not a ground truth. There is no labelled dataset of known bad prescribing records to calibrate against.
- The model only uses total spending and list size. Adding drug-category breakdowns — antibiotics, opioids, antidepressants — would almost certainly improve the quality of detections.
- Several NHS CCGs merged or reorganised between 2021 and 2026. Where a CCG's patient list suddenly doubles because of a merger, that will look like an anomaly even though nothing unusual is happening clinically.
- The analysis covers monthly totals, not individual prescriptions. A CCG could have one very unusual prescription hidden in otherwise normal totals — that would not be detected here.

Part 3 - Finding the Areas with the Greatest Unmet Need

1) What problem does this solve

Not all parts of England are equal when it comes to mental health services. Some areas are heavily deprived high unemployment, poor housing, low income and research consistently shows that deprivation is strongly linked to poorer mental health. The question this analysis tries to answer is: are the areas with the highest deprivation actually getting adequate mental health support through NHS Talking Therapies?

By mapping deprivation data alongside IAPT service performance data at ICB level, we can see where the gap between need and provision is largest. Those areas should arguably be the priority for additional investment or intervention.

2) Dataset

- **NHS Talking Therapies data**
  - The February 2026 monthly snapshot from NHS England, filtered to SubICB-level records. Five measures were extracted for each area: referrals received, people accessing services, recovery rate, average waiting time, and people waiting more than 90 days.
- **Index of Multiple Deprivation (IMD) 2025**
  - Published by the Department for Levelling Up, Housing and Communities, the IMD scores every small area in England across seven dimensions: income, employment, education, health, crime, housing, and environment. There are 33,755 LSOAs in England. For this analysis, scores were converted to a 0–100 scale where 100 means most deprived.
- **Geographic lookup and boundary files**
  - A lookup table was used to connect each LSOA to its parent ICB. A shapefile of ICB boundaries was then used to draw the maps, using the British National Grid (EPSG:27700) coordinate system.

3) How was the analysis done

1 **NHS data preparation.** The raw NHS file covers many different measure types. The five relevant measures were extracted, converted to numbers, and aggregated up to ICB level. An access rate was calculated as the percentage of referrals that went on to receive treatment.

2 **IMD preparation.** Column names were shortened and IMD ranks converted into an intuitive score running from 0 (least deprived) to 100 (most deprived).

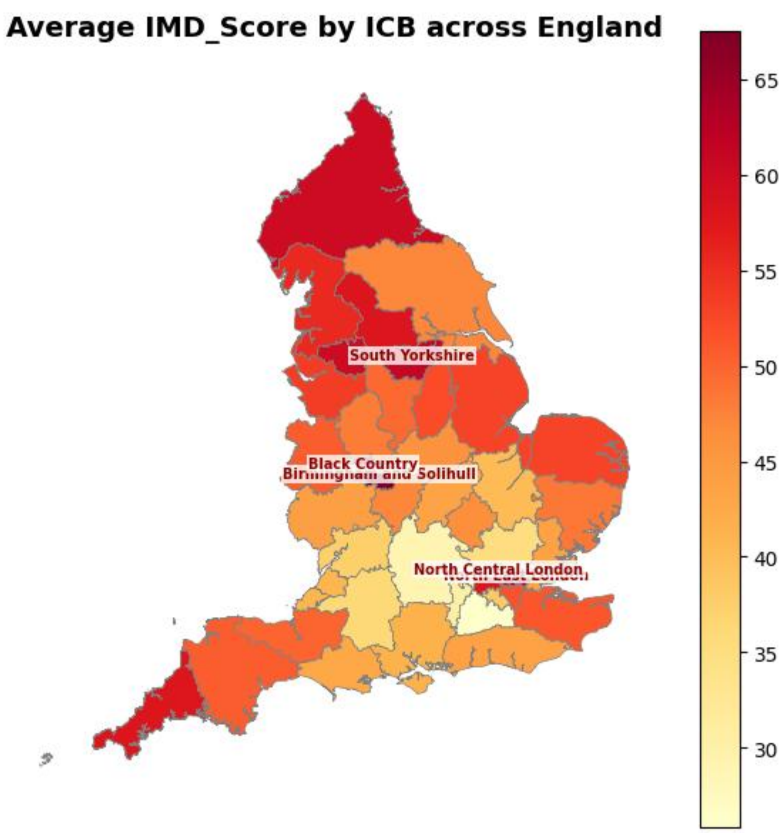
3 **Spatial join.** Each of the 33,755 LSOAs was matched to its ICB using the lookup table. IMD scores were then averaged up to ICB level, giving a single deprivation score per ICB. All 42 ICBs were successfully matched.

4 **Combining the datasets.** The NHS service metrics and IMD deprivation scores were merged on ICB name, creating one row per ICB with both sets of information.

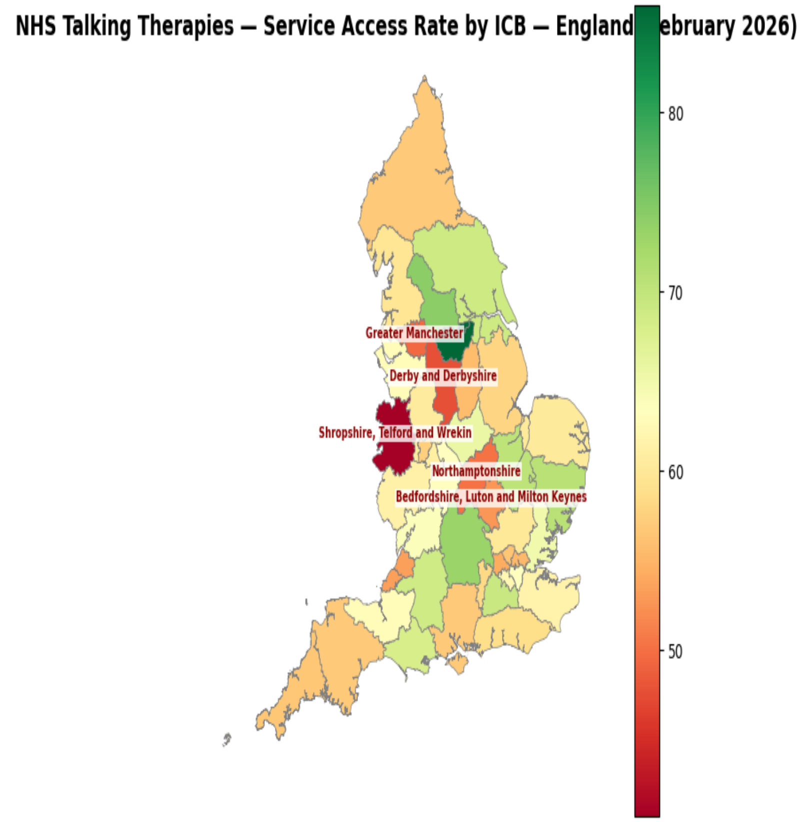
5 **Classifying underserved areas.** An ICB was flagged as high-need if its average deprivation score fell in the top 25% of all ICBs. It was flagged as low-service if its referrals, access rate, or recovery rate fell in the bottom 25%. ICBs that were both high-need and low-service were classified as underserved.

A composite "Unmet Need Score" was also calculated to rank the underserved ICBs by severity: 35% weight on deprivation, 30% on low access, 20% on low recovery rate, and 15% on long waiting times.

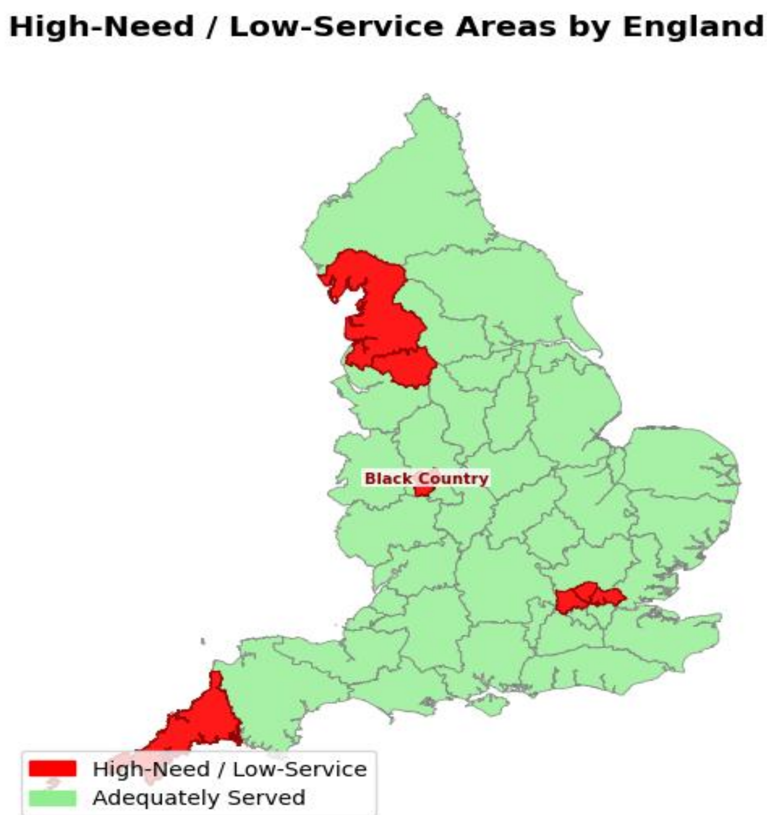
4) What do the maps show?



The deprivation picture broadly matches what we know about England's geography of inequality — higher deprivation in post-industrial northern towns, coastal communities, and some inner-London boroughs.



Comparing Figure 1 and Figure 2 side by side is where the story gets interesting. Some of the most deprived areas are also among the lowest for access — meaning the people who arguably need talking therapies most are least likely to actually get them.



5) Which areas need most attention

ICB	IMD score	Access rate	Avg wait (days)	Unmet need score
NHS Greater Manchester	60.1	49.5%	39.0	**74.4**
NHS Black Country	64.5	57.5%	49.2	**73.4**
NHS North East London	62.8	55.3%	32.5	62.7
NHS North Central London	60.2	56.4%	25.9	62.5
NHS North West London	57.7	54.2%	17.6	61.4
NHS Lancashire & South Cumbria	55.6	59.7%	25.8	60.3
NHS Cornwall & Isles of Scilly	57.8	56.6%	25.3	50.4
England average (all 42 ICBs)	—	**60.9%**	**26.5**	—

Greater Manchester has the highest unmet need score by some distance. It combines a high deprivation score with the worst access rate of any priority ICB (49.5%) and a notably long average wait of 39 days — 12.5 days above the England average.

The three London ICBs in the list (North East, North Central, North West London) represent a different challenge. Deprivation in these areas is high but driven by urban poverty concentrated in specific wards — the ICB average can mask very sharp within-borough inequalities.

An access rate below 60.9% (the England average) means that for every 10 people referred, fewer than 6 actually access treatment. For Greater Manchester, that figure is closer to 5 in 10. Closing this gap is where targeted investment would have the most impact.

6) Assumptions and honest review

- This is a February 2026 snapshot. A single month's data can be affected by seasonal patterns and may not represent the typical position.
- ICB-level averages hide variation within ICBs. Some districts within an ICB may be very well served while others are not.
- The 25th/75th percentile thresholds for defining "high-need" and "low-service" are reasonable starting points but are ultimately judgement calls. Moving the thresholds would change which ICBs appear on the priority list.
- The IMD scores reflect 2021 census boundaries. Since then, some areas will have seen economic changes that the scores do not yet capture.
- Access rate is a proxy. It does not tell us whether the treatment received was appropriate, whether people dropped out, or whether they recovered.

Overall Takeaways and Recommended

What this work tells us, in plain terms

Taken together, these three analyses paint a reasonably coherent picture:

- Demand for NHS Talking Therapies is broadly stable but seasonally variable, with a predictable winter dip and spring-to-autumn stability. Services can use the monthly forecasts to smooth out capacity planning.
- Some CCGs have been prescribing in ways that stand out consistently over multiple months — the same names appear on the anomaly list repeatedly, which suggests these are not one-off data errors and deserve clinical audit attention.
- There is a clear geographic mismatch between where deprivation is highest and where talking therapies services perform best. Seven ICBs stand out as priority areas where investment or targeted support is most warranted.

What should happen next?

For the forecasting model

- Rerun monthly as new IAPT data is published — a model trained on 38 months will improve as more data accumulates
- Build ICB-level forecasts, not just the England-wide total, so regional commissioners can plan independently
- Consider adding UK bank holidays and seasonal adjustment to reduce the November dip prediction error

For the anomaly detection

- Set up a review process: the anomaly CSV should go to a pharmacist or prescribing lead who can look at the top 20–30 flagged records each month
- Over time, collect feedback on which anomalies turned out to be genuine issues — this creates a labelled dataset that could train a better model
- When the API restriction is resolved, automate the monthly data pull so the analysis can run without manual downloads

For the geo-spatial analysis

- Run the same analysis on multiple months of IAPT data to see whether the priority ICBs are consistently underperforming or whether February 2026 was an unusual month for any of them
- Share the priority ICB list with NHS England's regional improvement teams to check whether targeted support is already in place
- Add a population denominator to produce true per-capita referral rates, which would give an even clearer picture of unmet need

Technical notes for analysts

- All code is in Python, using pandas, Prophet, statsmodels, scikit-learn, geopandas, and matplotlib. Notebooks are provided as supplementary files.
- The random seed for the Isolation Forest is fixed at 42 so results are reproducible.
- The spatial analysis requires GDAL/GEOS to be installed — these are not always included in default Python environments.
- Data sources: NHS Talking Therapies monthly statistics (NHS England), IoD 2025 (DLUHC), OpenPrescribing API (Bennett Institute for Health Policy), ONS geography boundary files. All are publicly available.